

House Price Prediction – Project Summary

Overview

This project focused on developing machine learning models to predict house prices using a housing dataset containing property characteristics such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and other amenities. The objective was to analyze the factors affecting house prices and evaluate the performance of different regression models for accurate price prediction.

Methodology

The dataset was loaded and explored using Pandas. Data preprocessing included checking for missing values, removing duplicate records, and converting categorical variables into numerical format using one-hot encoding. The data was then divided into training and testing sets using an 80:20 split.

Two machine learning models were implemented and compared:

- Linear Regression
- Random Forest Regressor

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score.

Results

The Linear Regression model achieved the best performance with an R² score of approximately **0.65**, indicating that it was able to explain about 65% of the variation in house prices. Surprisingly, Linear Regression outperformed the Random Forest model on this dataset, suggesting that the relationship between the housing features and property prices is largely linear.

Performance comparison:

Model	R ² Score
Linear Regression	0.653
Random Forest Regressor	0.612

Key Findings

Feature importance and correlation analysis revealed that **area**, **bathrooms**, and **parking availability** were among the most influential factors affecting house prices. Larger properties with better amenities generally commanded higher market values. The analysis also showed strong positive relationships between property size and overall price.

Business Recommendation

Real estate agencies and property developers should prioritize factors such as property size, bathroom count, parking facilities, and premium amenities when estimating property values. These characteristics demonstrated the strongest influence on house prices and can help improve pricing accuracy and marketing strategies.

Conclusion

The project successfully demonstrated the use of machine learning techniques for house price prediction. Linear Regression proved to be the most effective model for this dataset, while feature importance analysis provided valuable insights into the factors influencing property values. The findings can assist real estate businesses in making more informed pricing decisions and understanding key market drivers.